

Laplace-tilted importance sampling for finite- T cluster spin DMFT

Notes on the implementation in `CspinDMFT_finite_T`

1 Setting and the failure mode

The self-consistent Monte Carlo loop draws a mean-field noise trajectory $V = \{V_i^\alpha(\tau)\}_{i,\alpha,\tau}$ from a multivariate Gaussian

$$p(V) = \mathcal{N}(\mu, \Sigma_V), \quad (1)$$

where μ is the (static) mean field and Σ_V is the mean-field covariance built from the previous iteration's spin correlations (the Frequency-domain block-diagonal structure already implemented by `FrequencyCovarianceCluster`). For each sampled V we compute the imaginary-time-ordered propagator and its trace

$$Z(V) = \text{Tr } T_\tau \exp\left(-\int_0^\beta H[V(\tau)] d\tau\right), \quad H[V] = J \mathbf{S} \cdot \mathbf{S} + V \cdot \mathbf{S}. \quad (2)$$

The observable of interest is

$$\langle \mathcal{O} \rangle = \frac{\mathbb{E}_p[Z(V) \mathcal{O}(V)]}{\mathbb{E}_p[Z(V)]}, \quad \mathcal{O}(V) \equiv \frac{\text{Tr}[F(V) S B(V) S]}{Z(V)}, \quad (3)$$

estimated by the self-normalised Monte Carlo ratio $\widehat{\langle \mathcal{O} \rangle} = \sum_s \text{Tr}[F S B S]_s / \sum_s Z_s$ with $V_s \sim p$.

The effective sample size collapses with β . The variance of the estimator is controlled by the spread of the importance weights $w_s = Z(V_s)$ under p . A useful diagnostic is the effective sample size

$$\text{ESS} \equiv \frac{(\sum_s w_s)^2}{\sum_s w_s^2} \in [1, M]. \quad (4)$$

For a single spin in field V with $Z(V) = 2 \cosh(\beta|V|/2)$ and $V \sim \mathcal{N}(0, \sigma^2)$,

$$\frac{\text{Var}_p[Z]}{\mathbb{E}_p[Z]^2} \approx \exp\left(\frac{\beta^2 \sigma^2}{4}\right) - 1, \quad (5)$$

so $\text{ESS}/M \sim \exp(-\beta^2 \sigma^2/4)$. For the cluster the exponent picks up an additive factor in the number of sites, so ESS/M degrades *exponentially* in both β and N . This is the origin of the “huge stds at low T ” failure mode and of the overflow NaNs observed in the `--uncoupled_spins` branch.

2 The ideal proposal and why it is intractable

For any positive proposal $q(V) > 0$,

$$\langle \mathcal{O} \rangle = \frac{\mathbb{E}_q[w(V) Z(V) \mathcal{O}(V)]}{\mathbb{E}_q[w(V) Z(V)]}, \quad w(V) \equiv \frac{p(V)}{q(V)}. \quad (6)$$

The zero-variance choice is $q^*(V) \propto p(V) Z(V)$, for which $w(V) Z(V) \equiv \text{const}$ and the estimator becomes the plain mean of $\mathcal{O}$ over $V_s \sim q^*$. We cannot sample from q^* directly because $Z(V)$ is the partition function of a non-trivial imaginary-time-ordered product, with no closed form.

3 Laplace approximation to q^*

We expand $\log p(V) + \log Z(V)$ to second order around its joint maximum V_* , giving the Gaussian proposal

$$q(V) = \mathcal{N}(V_*, H^{-1}), \quad H = \Sigma_V^{-1} - \chi_V[V_*], \quad (7)$$

where

$$V_* \text{ satisfies } \Sigma_V^{-1} V_* = \langle \mathbf{S} \rangle_{V_*}, \quad (8)$$

$$\chi_V[V](\tau, \tau')_{i\alpha, j\beta} \equiv \frac{\partial^2 \log Z[V]}{\partial V_i^\alpha(\tau) \partial V_j^\beta(\tau')}. \quad (9)$$

The DMFT self-consistency *already* solves (??) for the mean, so $V_* = \mu$ for free. Only the Gaussian *covariance* changes: it becomes wider than Σ_V by the susceptibility correction.

3.1 Continuum and discrete forms of χ_V

For a static field V , $\partial \log Z / \partial V_i^\alpha = -\beta \langle S_i^\alpha \rangle_V$. For an imaginary-time-dependent field one finds, in the continuous formulation,

$$\frac{\partial^2 \log Z[V]}{\partial V_i^\alpha(\tau) \partial V_j^\beta(\tau')} = \chi_S(\tau, \tau')_{i\alpha, j\beta}, \quad \chi_S \equiv \left\langle T_\tau S_i^\alpha(\tau) S_j^\beta(\tau') \right\rangle_{\text{conn}}. \quad (10)$$

On the discrete time grid $\tau_n = n \Delta\tau$, $\Delta\tau = \beta/N_t$, where $\partial V_i^\alpha(\tau_n)$ acts as if integrated against $\Delta\tau$,

$$\chi_V|_{\text{disc}} = (\Delta\tau)^2 \chi_S, \quad (11)$$

which gives the implementation factor `tilt_scale = dt*dt`. With `tilt_scale = 1` the quadratic form in the weight overflows within the first sample; with `tilt_scale =  $\Delta\tau^2$` the ESS plateau increases monotonically (§??).

3.2 Block-wise Matsubara implementation

Σ_V and χ_V are translationally invariant in imaginary time, hence block-diagonal in the cosine/sine Matsubara basis used by `FrequencyCovarianceCluster`. For each frequency n and symmetry block b ,

$$H_n^{(b)} = [\Sigma_V^{(b)}_n]^{-1} - \text{tilt_scale} \chi_S^{(b)}_n, \quad \Sigma_V^{\text{tilted}(b)} = [H_n^{(b)}]^{-1}. \quad (12)$$

If any block fails the positive-definiteness check $\lambda_{\min}(H_n^{(b)}) \leq 0$ we keep the untilted block for that frequency. This robust fallback matters near a phase transition where χ can become large enough to invert the sign of an eigenvalue. The cosine-Fourier convention used to build χ_n matches the one used by `fill()`, so the quadratic form below is consistent without an extra normalisation factor.

4 Per-sample weight

Because p and q share the mean $\mu = V_*$,

$$\begin{aligned} \log w(V) &= \log p(V) - \log q(V) \\ &= -\frac{1}{2} (V - \mu)^\top [\Sigma_V^{-1} - H] (V - \mu) - \frac{1}{2} \log \det(\Sigma_V H) \\ &= -\frac{1}{2} \text{tilt_scale} (V - \mu)^\top \chi_S (V - \mu) + C, \end{aligned} \quad (13)$$

with C a sample-independent constant that drops out of any self-normalised estimator. In the orthonormal cosine/sine basis used to draw V , the quadratic form decouples across Matsubara frequencies,

$$(V - \mu)^\top \chi_S (V - \mu) = \sum_n \tilde{V}_n^\top \chi_{S n} \tilde{V}_n, \quad (14)$$

which is what `compute_log_pq_weight()` evaluates after a forward Fourier of the sampled time-domain trajectory.

Why the sign is correct. Tail samples of V have anomalously large $Z(V)$. The optimal q^* puts more probability there, so when we draw from q those samples are oversampled relative to the target p . The compensating weight $w = p/q$ must therefore be *smaller* in the tail — exactly the behaviour of (??) for `tilt_scale` > 0.

5 What enters the running accumulators

Every per-sample quantity that the naïve code multiplies by Z gets replaced by the product wZ , and every per-sample quantity multiplied by an unnormalised trace $x = \text{Tr}[FSBS]$ by $w x$:

$$\text{partition function: } \sum_s Z_s \longrightarrow \sum_s w_s Z_s, \quad (15)$$

$$\text{correlator estimator: } \sum_s x_s \longrightarrow \sum_s w_s x_s, \quad (16)$$

$$\text{Delta-method second moments: } \sum_s x_s^2, x_s Z_s, Z_s^2 \longrightarrow \sum_s (w_s x_s)^2, (w_s x_s)(w_s Z_s), (w_s Z_s)^2, \quad (17)$$

which preserves the Delta-method formula for the std of the ratio estimator and the ESS formula (??) with $w_s Z_s$ in place of Z_s . In the implementation a single argument `weight` to `compute_spin_observables()` applies these substitutions consistently across all accumulators.

6 Diagnostics and convergence

For the $N = 2$, $J = 0.5$ square cluster at $M = 4 \times 10^4$ samples per DMFT iteration, the converged ESS plateaus are

β	naïve ESS/ M	tilted ESS/ M	variance reduction
1	0.994	1.000	$\sim 1\times$
3	0.751	0.947	$\sim 1.3\times$
5	0.341	0.894	$\sim 2.6\times$

Mean correlators agree within sample stds at all τ , all β (`Plots/tilted_vs_naive_N2_J0.5.{pdf,png}`), confirming that reweighting is unbiased and that the variance reduction is genuine.

7 Open issues

1. **Susceptibility from spin correlations.** The implementation currently builds χ_S from the previous iteration's `my_spin_correlations_Re` with the disconnected $\langle S \rangle \langle S \rangle$ subtracted. This is the cluster susceptibility of the *bath-dressed* impurity, not the susceptibility at the static saddle $V_\star = \mu$. They agree at self-consistency and differ away from it, which is why ESS/ M at $\beta = 5$ plateaus near 0.9 rather than 1.0. A deterministic propagator run at zero noise would yield the Laplace-exact χ .

2. **Reduced layout for χ .** The `correlation_categories` layout stores only representative pairs; `FrequencyCovarianceCluster::fill` requires an all-pairs `CluCorrTen`. The current `build_connected_susceptibility` reconstructs missing pairs by translation fallback. For non-translationally-invariant problems (open boundaries, disorder) a proper mapping is required.
3. **Positive definiteness near criticality.** The block-wise PD check guards against $\chi_{S_n}^{(b)}$ exceeding $[\Sigma_V^{(b)}]^{-1}$ in any direction. At a phase transition this guard activates for the dominant mode, the tilt is suppressed there, and ESS recovery degrades smoothly rather than blowing up. A monitoring counter (number of untilted blocks per iteration) is worth promoting to a stored HDF5 diagnostic.
4. **Uncoupled branch.** `--tilted_is` is currently gated on `mypspace.uncoupledspins!`. *Extending it to the uncoupled sites susceptibility is analytic and would address the overflow NaN at $\beta \gtrsim 5$ at its statistical source rather than via log-space bookkeeping.*
5. **Compounding with cluster size.** The relative-variance bound (??) adds in the number of sites. The $N = 2$ improvement from 0.34 to 0.89 at $\beta = 5$ would compound to roughly $(0.89/0.34)^{N/2}$ across N sites if the tilt remains effective per site. This is what makes the Laplace tilt structurally important for production-scale $N = 9$ runs.